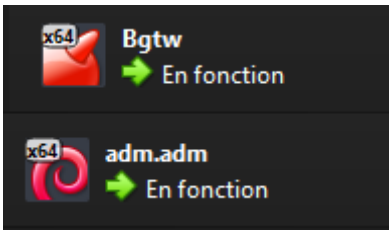
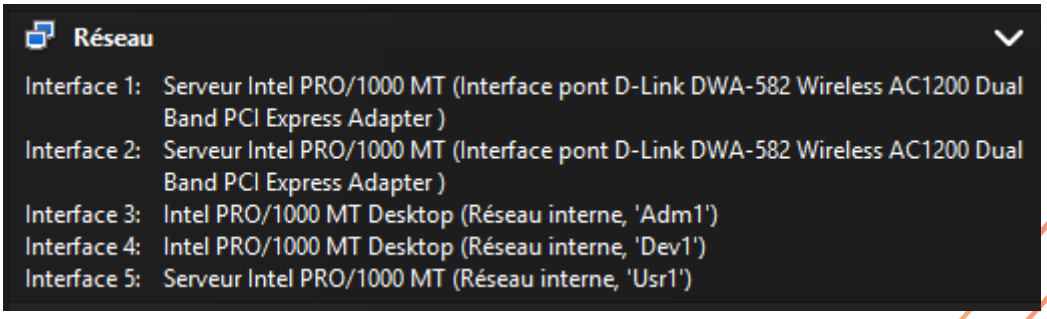
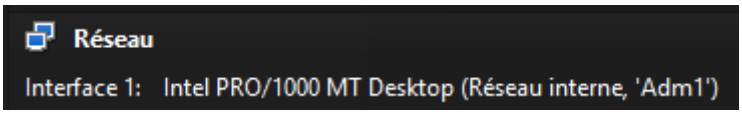
Rendu AP DMZ

SALIM Salim SIO2A

Installation des machines virtuelles :

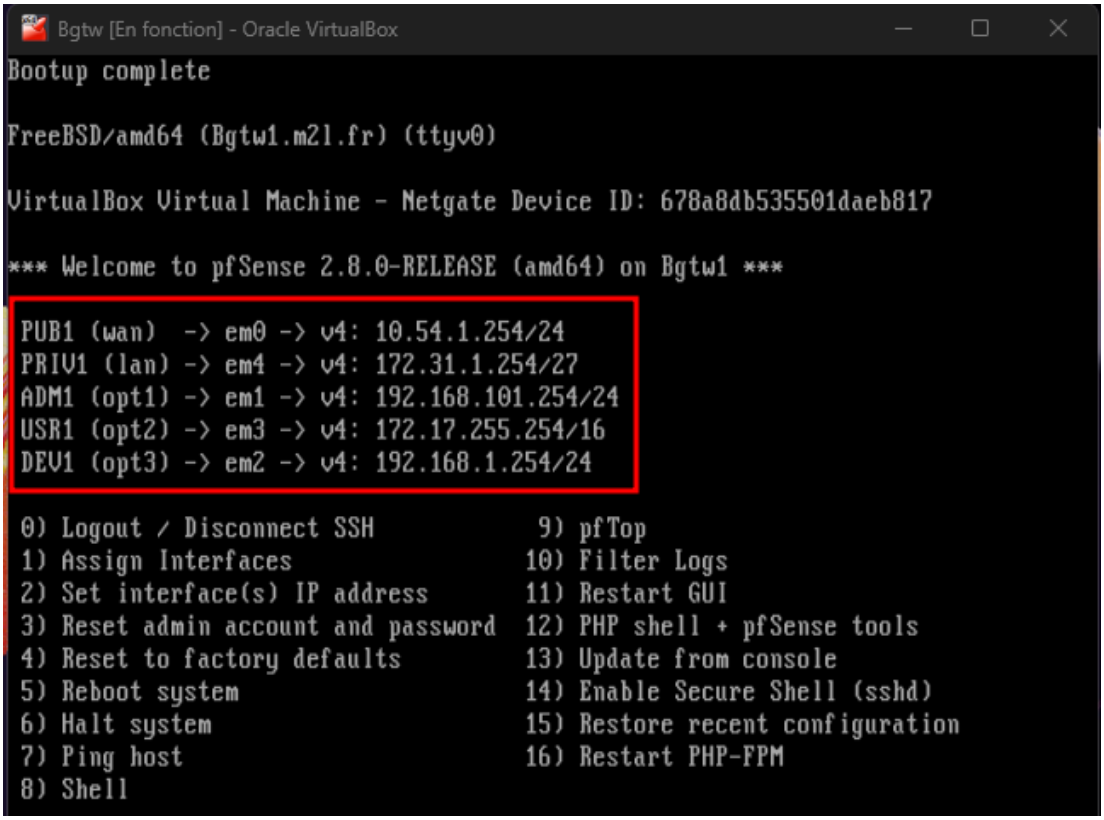






Les cartes réseaux sont différent car je réalise les rendus chez moi.

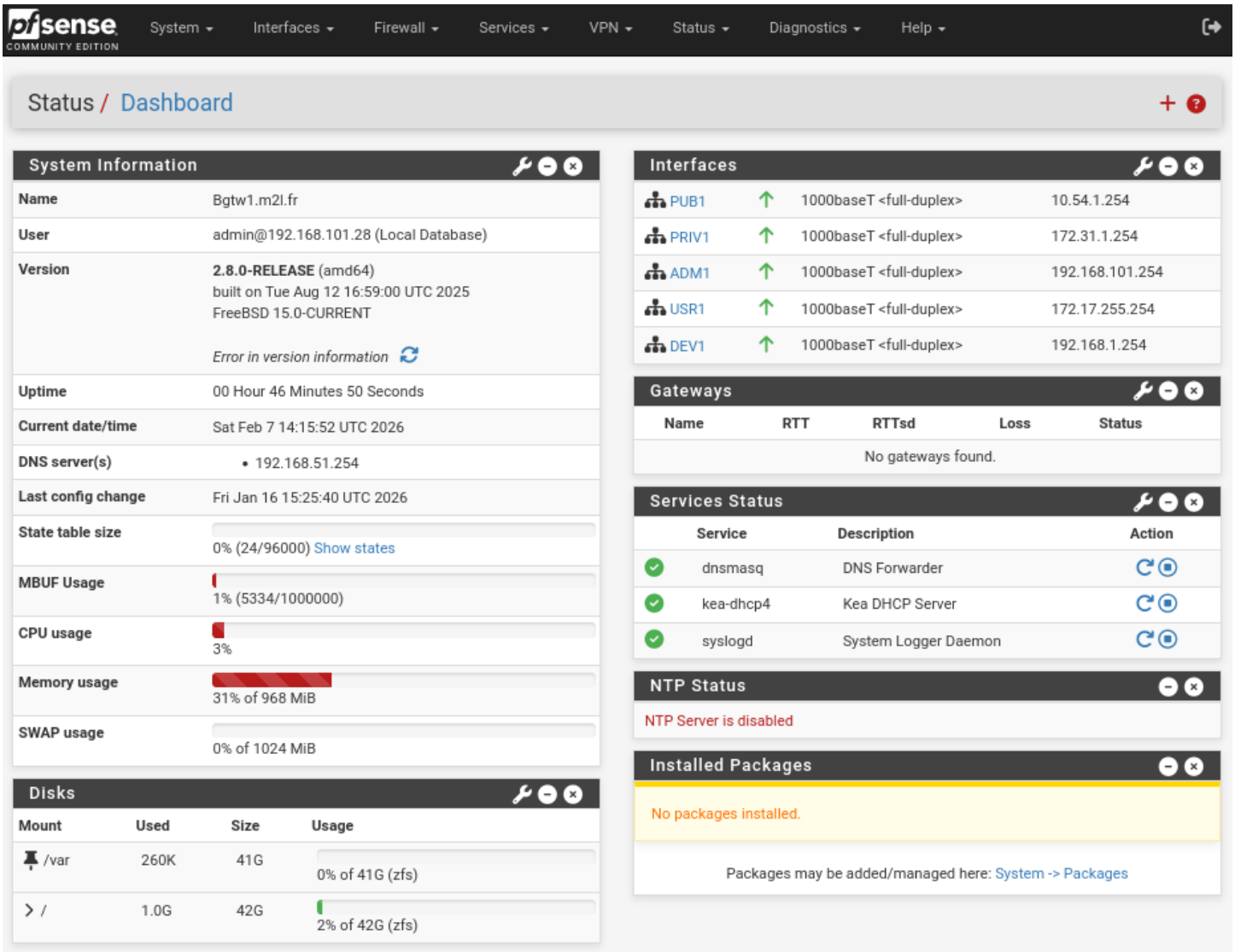
Mise en place du routeur Bgtw :



D'abord nous avons ajouté les cartes réseaux 5 au total. Pour mettre en place le 5ème on devait utiliser un programme qui nous été fourni.

Il fallait faire attention à ne pas se tromper d'adresses IP pendant l'assignement. Sinon ça ne fonctionnera pas. Ensuite on utilise un Kali pour l'administrer.

Dashboard :



Status.Interfaces :

Status / Interfaces

PUB1 Interface (wan, em0)

Status	up
MAC Address	08:00:27:d9:39:24
IPv4 Address	10.54.1.254
Subnet mask IPv4	255.255.255.0
IPv6 Link Local	fe80::a00:27ff:fed9:3924%em0
MTU	1500
Media	1000baseT <full-duplex>
In/out packets	502/0 (125 KiB/0 B)
In/out packets (pass)	502/0 (125 KiB/0 B)
In/out packets (block)	28/0 (2 KiB/0 B)
In/out errors	0/0
Collisions	0

PRIV1 Interface (lan, em4)

Status	up
MAC Address	08:00:27:da:d0:af
IPv4 Address	172.31.1.254
Subnet mask IPv4	255.255.255.224
IPv6 Link Local	fe80::a00:27ff:feda:d0af%em4
MTU	1500
Media	1000baseT <full-duplex>
In/out packets	1424/0 (333 KiB/0 B)
In/out packets (pass)	1424/0 (333 KiB/0 B)
In/out packets (block)	62/0 (5 KiB/0 B)
In/out errors	0/0
Collisions	0

ADM1 Interface (opt1, em1)

Status	up
MAC Address	08:00:27:65:87:dd
IPv4 Address	192.168.101.254
Subnet mask IPv4	255.255.255.0
IPv6 Link Local	fe80::a00:27ff:fe65:87dd%em1
MTU	1500
Media	1000baseT <full-duplex>
In/out packets	632/885 (150 KiB/668 KiB)
In/out packets (pass)	632/885 (150 KiB/668 KiB)
In/out packets (block)	0/1 (0 B/88 B)
In/out errors	0/0
Collisions	0

USR1 Interface (opt2, em3)

Status	up
MAC Address	08:00:27:01:2f:de
IPv4 Address	172.17.255.254
Subnet mask IPv4	255.255.0.0
IPv6 Link Local	fe80::a00:27ff:fe01:2fde%em3
MTU	1500
Media	1000baseT <full-duplex>
In/out packets	0/0 (0 B/0 B)
In/out packets (pass)	0/0 (0 B/0 B)
In/out packets (block)	0/0 (0 B/0 B)
In/out errors	0/0
Collisions	0

DEV1 Interface (opt3, em2)

Status	up
MAC Address	08:00:27:1f:4b:e0
IPv4 Address	192.168.1.254
Subnet mask IPv4	255.255.255.0
IPv6 Link Local	fe80::a00:27ff:fe1f:4be0%em2
MTU	1500
Media	1000baseT <full-duplex>
In/out packets	0/0 (0 B/0 B)
In/out packets (pass)	0/0 (0 B/0 B)
In/out packets (block)	0/5 (0 B/384 B)
In/out errors	0/0
Collisions	0

Voici les 5 cartes réseaux. Avec toutes les informations nécessaires.

Status.Services :

Status / Services

Services

Service	Description	Status	Actions
dnsmasq	DNS Forwarder	✓	
kea-dhcp4	Kea DHCP Server	✓	
syslogd	System Logger Daemon	✓	

Service.Dns-forwarder :

Services / DNS Forwarder

General DNS Forwarder Options

Enable ☑ Enable DNS forwarder

Ignore System DNS ☐ Do not use system DNS servers

If this option is set the configured system DNS servers will be ignored and custom "server=" options must be used.

DNS Query Forwarding

☐ Query DNS servers sequentially

If this option is set pfSense DNS Forwarder (dnsmasq) will query the DNS servers sequentially in the order specified (*System - General Setup - DNS Servers*), rather than all at once in parallel.

☐ Require domain

If this option is set pfSense DNS Forwarder (dnsmasq) will not forward A or AAAA queries for plain names, without dots or domain parts, to upstream name servers. If the name is not known from /etc/hosts or DHCP then a "not found" answer is returned.

☐ Do not forward private reverse lookups

If this option is set pfSense DNS Forwarder (dnsmasq) will not forward reverse DNS lookups (PTR) for private addresses (RFC 1918) to upstream name servers. Any entries in the Domain Overrides section forwarding private "n.n.n.in-addr.arpa" names to a specific server are still forwarded. If the IP to name is not known from /etc/hosts, DHCP or a specific domain override then a "not found" answer is immediately returned.

Listen Port 53

The port used for responding to DNS queries. It should normally be left blank unless another service needs to bind to TCP/UDP port 53.

Interfaces

ADM1
USR1
DEV1

Interface IPs used by the DNS Forwarder for responding to queries from clients. If an interface has both IPv4 and IPv6 IPs, both are used. Queries to other interface IPs not selected above are discarded. The default behavior is to respond to queries on every available IPv4 and IPv6 address.

Strict binding ☐ Strict interface binding

If this option is set, the DNS forwarder will only bind to the interfaces containing the IP addresses selected above, rather than binding to all interfaces and discarding queries to other addresses.

This option does NOT work with IPv6. If set, dnsmasq will not bind to IPv6 addresses.

Il faut l'activer et choisir les interfaces dont on veut l'appliquer.

Diagnostic.Routes :

IPv4 Routes

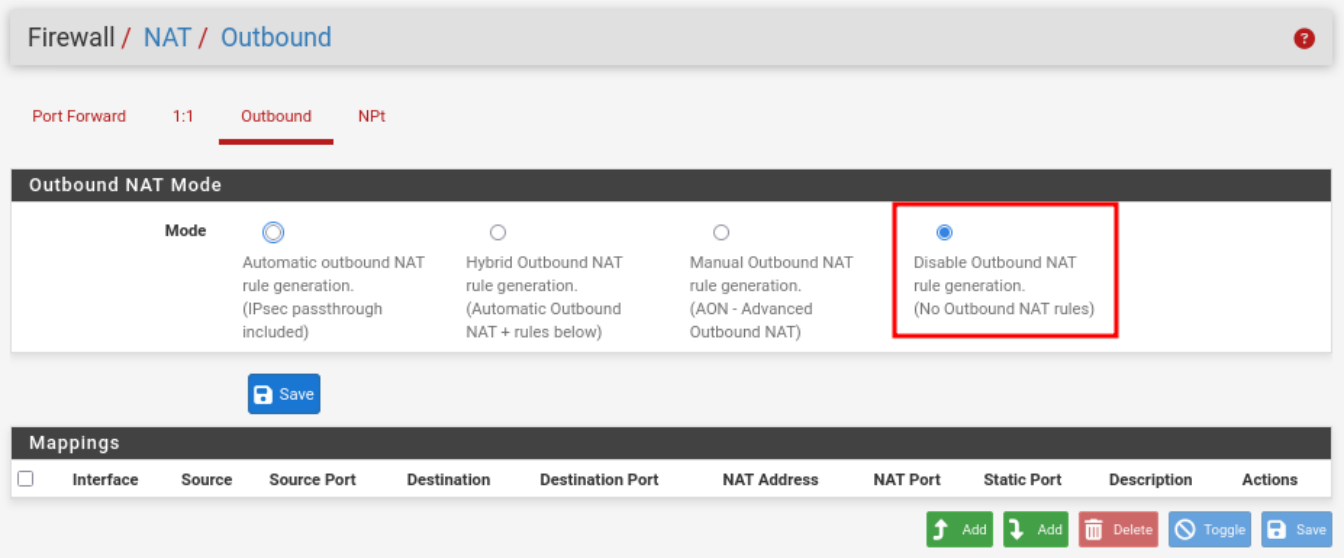
Destination	Gateway	Flags	Uses	MTU	Interface	Expire
10.54.1.0/24	link#1	U	1	1500	em0	
10.54.1.254	link#7	UHS	3	16384	lo0	
127.0.0.1	link#7	UH	2	16384	lo0	
172.17.0.0/16	link#4	U	8	1500	em3	
172.17.255.254	link#7	UHS	9	16384	lo0	
172.31.1.224/27	link#5	U	4	1500	em4	
172.31.1.254	link#7	UHS	5	16384	lo0	
192.168.1.0/24	link#3	U	10	1500	em2	
192.168.1.254	link#7	UHS	11	16384	lo0	
192.168.101.0/24	link#2	U	6	1500	em1	
192.168.101.254	link#7	UHS	7	16384	lo0	

Diagnostics.Sockets :

IPv4 System Socket Information

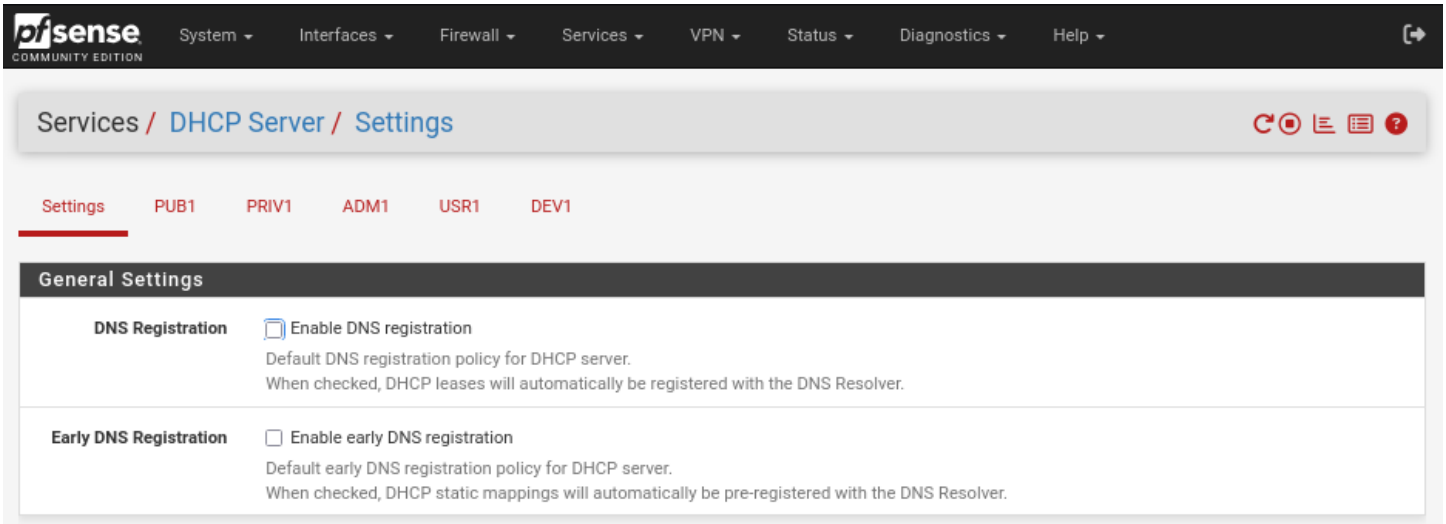
USER	COMMAND	PID	FD	PROTO	LOCAL	FOREIGN
root	php-fpm	65015	4	udp4	*:*	*:*
root	kea-dhcp4	82610	16	udp4	192.168.101.254:67	*:*
root	kea-dhcp4	82610	18	udp4	192.168.1.254:67	*:*
root	kea-dhcp4	82610	20	udp4	172.17.255.254:67	*:*
root	php-fpm	37290	4	udp4	*:*	*:*
root	syslogd	93711	10	udp4	*:514	*:*
root	nginx	56945	5	tcp4	*:80	*:*
root	nginx	56828	5	tcp4	*:80	*:*
root	nginx	56540	5	tcp4	*:80	*:*
nobody	dnsmasq	11105	4	udp4	*:53	*:*
nobody	dnsmasq	11105	5	tcp4	*:53	*:*
root	php-fpm	428	4	udp4	*:*	*:*
root	php-fpm	427	4	udp4	*:*	*:*
root	php-fpm	426	4	udp4	*:*	*:*

Firewall.nat.port-forward :



Il faut désactiver le firewall outbound du côté Back Gateaway.

Services.dhcp-server :



Pub I :

Primary Address Pool

Subnet	10.54.1.0/24
Subnet Range	10.54.1.1 - 10.54.1.254
Address Pool Range	From: ; To: The specified range for this pool must not be within the range configured on any other address pool for this interface.
Additional Pools	+ Add Address Pool If additional pools of addresses are needed inside of this subnet outside the above range, they may be specified here.

Priv 1 :

Primary Address Pool

Subnet	172.31.1.224/27
Subnet Range	172.31.1.225 - 172.31.1.254
Address Pool Range	From: 192.168.1.100 ; To: 192.168.1.199 The specified range for this pool must not be within the range configured on any other address pool for this interface.
Additional Pools	+ Add Address Pool If additional pools of addresses are needed inside of this subnet outside the above range, they may be specified here.

Adm 1 :

Primary Address Pool

Subnet	192.168.101.0/24
Subnet Range	192.168.101.1 - 192.168.101.254
Address Pool Range	From: 192.168.101.10 ; To: 192.168.101.200 The specified range for this pool must not be within the range configured on any other address pool for this interface.
Additional Pools	+ Add Address Pool If additional pools of addresses are needed inside of this subnet outside the above range, they may be specified here.

Usr 1 :

Primary Address Pool

Subnet	172.17.0.0/16
Subnet Range	172.17.0.1 - 172.17.255.254
Address Pool Range	From: 172.17.1.10 ; To: 172.17.1.200 The specified range for this pool must not be within the range configured on any other address pool for this interface.
Additional Pools	+ Add Address Pool If additional pools of addresses are needed inside of this subnet outside the above range, they may be specified here.

Dev 1 :

Primary Address Pool

Subnet	192.168.1.0/24
Subnet Range	192.168.1.1 - 192.168.1.254
Address Pool Range	192.168.1.10 (From) — 192.168.1.200 (To)
	The specified range for this pool must not be within the range configured on any other address pool for this interface.
Additional Pools	+ Add Address Pool
	If additional pools of addresses are needed inside of this subnet outside the above range, they may be specified here.

Voici toutes les adresses IP et les plages de tous les réseaux.

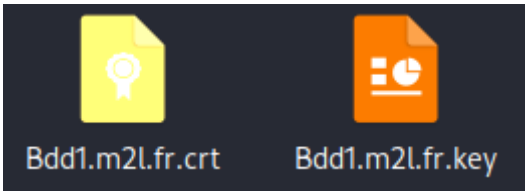
System.Certificats :

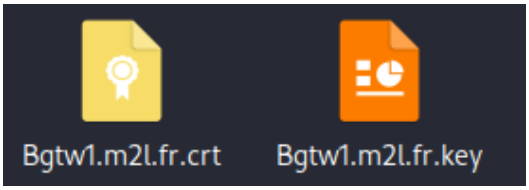
Bgtw1.m2l.fr *Server Certificate* CA: **No** Server: **Yes**	M2L-CA	CN=Bgtw1.m2l.fr Valid From: **Sun, 14 Dec 2025 13:17:41 +0000** Valid Until: **Wed, 12 Dec 2035 13:17:41 +0000**	
Fgtw1.m2l.fr *Server Certificate* CA: **No** Server: **Yes**	M2L-CA	CN=Fgtw1.m2l.fr Valid From: **Sun, 14 Dec 2025 13:21:17 +0000** Valid Until: **Wed, 12 Dec 2035 13:21:17 +0000**	
Bdd1.m2l.fr *Server Certificate* CA: **No** Server: **Yes**	M2L-CA	CN=Bdd1.m2l.fr Valid From: **Sun, 14 Dec 2025 13:23:17 +0000** Valid Until: **Wed, 12 Dec 2035 13:23:17 +0000**	
Web1.m2l.fr *Server Certificate* CA: **No** Server: **Yes**	M2L-CA	CN=Web1.m2l.fr Valid From: **Sun, 14 Dec 2025 13:24:53 +0000** Valid Until: **Wed, 12 Dec 2035 13:24:53 +0000**	
adm.vpna *User Certificate* CA: **No** Server: **No**	M2L-CA	CN=adm.vpna Valid From: **Sun, 14 Dec 2025 13:27:09 +0000** Valid Until: **Wed, 12 Dec 2035 13:27:09 +0000**	
dev.vpnd *User Certificate* CA: **No** Server: **No**	M2L-CA	CN=dev.vpnd Valid From: **Sun, 14 Dec 2025 13:27:36 +0000** Valid Until: **Wed, 12 Dec 2035 13:27:36 +0000**	

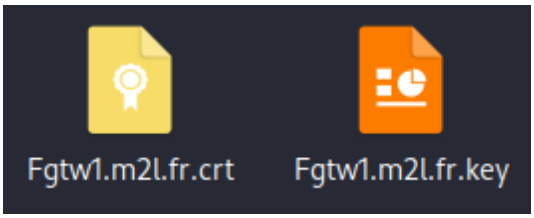
Et certificat d'autorité,

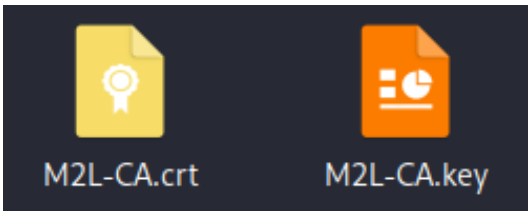
M2L-CA	✓	*self-signed*	6	CN=internal-ca Valid From: **Sun, 14 Dec 2025 13:12:21 +0000** Valid Until: **Wed, 12 Dec 2035 13:12:21 +0000**	

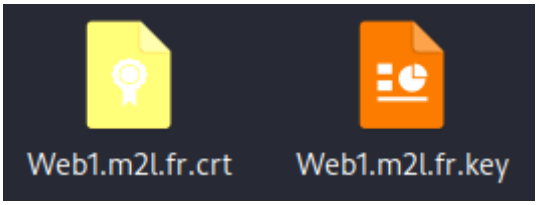
Et voici tous les certificats :

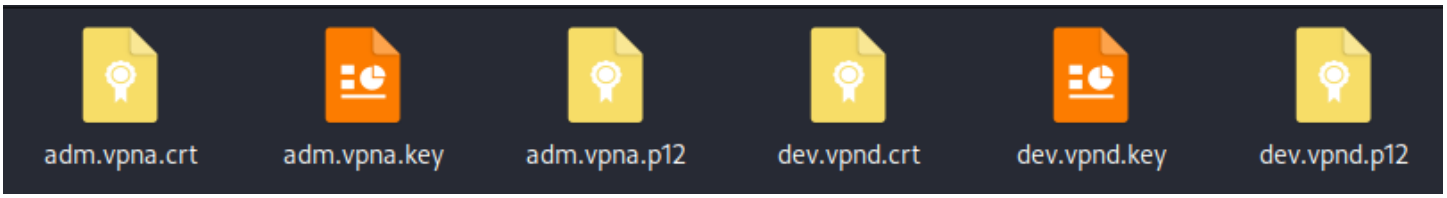












Firewall.rules :

Pub 1 :

Firewall / Rules / PUB1

Floating PUB1 PRIV1 ADM1 USR1 DEV1

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
✓	8/444 KiB	IPv4 *	*	*	*	*	*	none		TOUT PASSE	

Add Add Delete Toggle Copy Save Separator

Priv 1 :

Firewall / Rules / PRIV1

Floating PUB1 PRIV1 ADM1 USR1 DEV1

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
✓	0/1 KiB	*	*	*	PRIV1 Address	80	*	*		Anti-Lockout Rule	
✓	9/244 KiB	IPv4 *	*	*	*	*	*	none		TOUT PASSE	

Add Add Delete Toggle Copy Save Separator

Adm 1 :

Firewall / Rules / ADM1

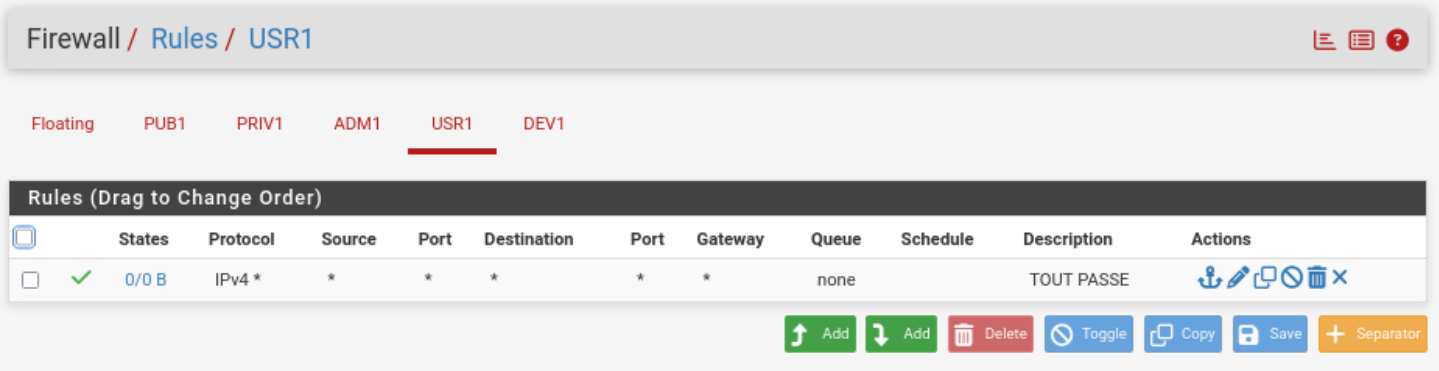
Floating PUB1 PRIV1 ADM1 USR1 DEV1

Rules (Drag to Change Order)

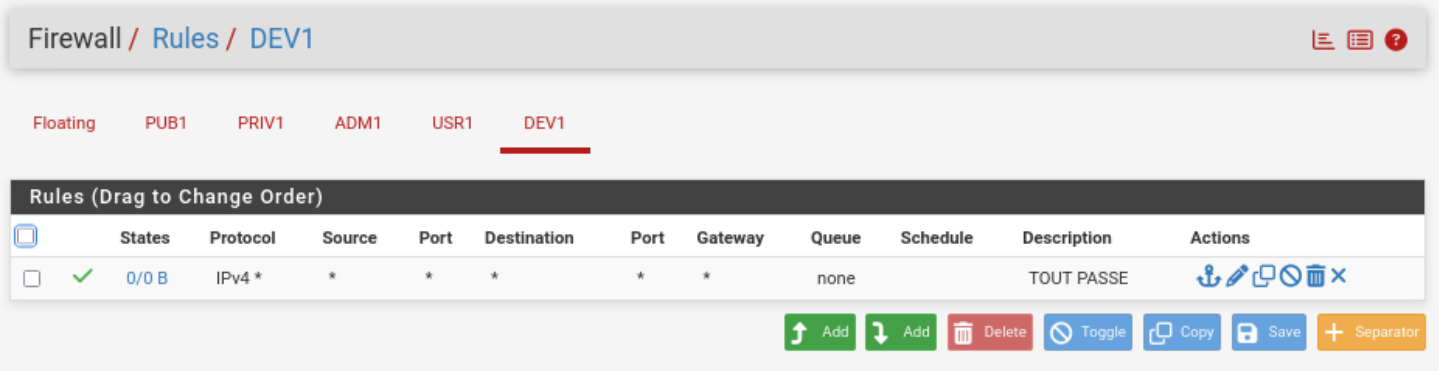
	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
✓	1/1.15 MiB	IPv4 *	*	*	*	*	*	none		TOUT PASSE	

Add Add Delete Toggle Copy Save Separator

Usr 1 :


Firewall / Rules / USR1
Floating
PUB1
PRIV1
ADM1
USR1
DEV1
Rules (Drag to Change Order)
States
Protocol
Source
Port
Destination
Port
Gateway
Queue
Schedule
Description
Actions
0/0 B
IPv4 *
*
*
*
*
*
none
TOUT PASSE
Add
Add
Delete
Toggle
Copy
Save
Separator

Dev 1 :


Firewall / Rules / DEV1
Floating
PUB1
PRIV1
ADM1
USR1
DEV1
Rules (Drag to Change Order)
States
Protocol
Source
Port
Destination
Port
Gateway
Queue
Schedule
Description
Actions
0/0 B
IPv4 *
*
*
*
*
*
none
TOUT PASSE
Add
Add
Delete
Toggle
Copy
Save
Separator